



Iterative Solve

Find the value of one variable that makes a result -- a variable, a mass or a volume -- hit a target, by re-running a list of features; keep only the answer.

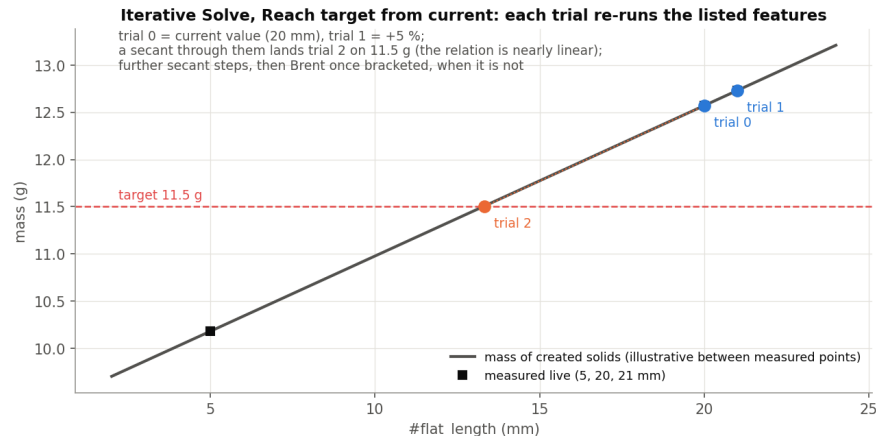


What it does

Setting one number so a result comes out right usually means a Pattern 'Reapply features' trick: slow (about 6 % closer per pass) and it keeps every pass. Iterative Solve treats the listed features as a function of the variable, solves for the target and keeps only the accepted trial.

Use it when

- A dimension must give a target mass or volume
- A variable must make another variable hit a value
- Find the first value that meets a condition



Stray_Weight: #flat_length for a target mass -- trial 2 lands on it.



Trials

each trial: set the variable, re-run every listed feature under its own id, measure

originals

trial 0 $x = 20$

trial 1 $x = 21$

trial 2 $x = 13.3$

the listed features: deleted once a trial is accepted

rejected: bodies deleted

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accepted: kept, #flat_length = 13.3

One trial

Set the variable, re-run every listed feature under its own id, measure.

Keep the answer

Rejected trials are deleted; the accepted one is kept, the originals deleted, the variable set.

No answer

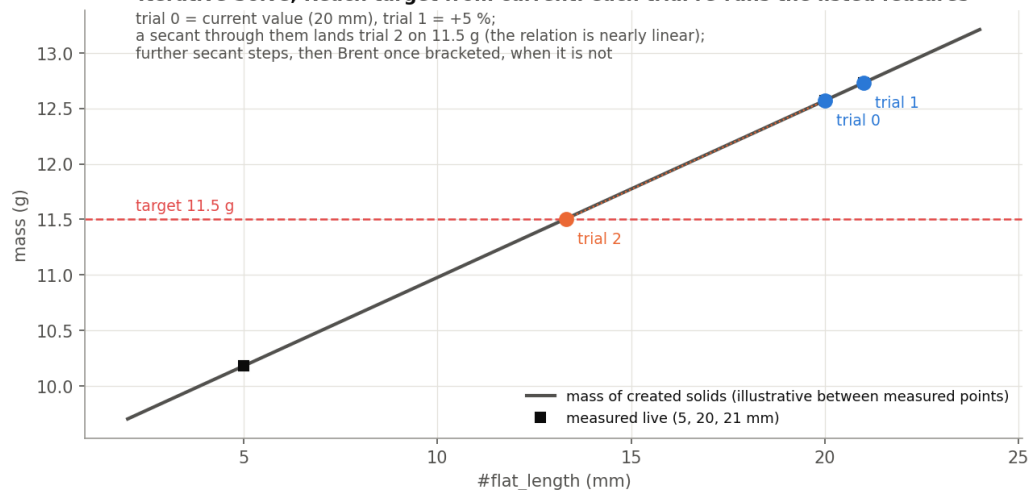
An error rolls the whole feature back; the originals stay.



Methods

Iterative Solve, Reach target from current: each trial re-runs the listed features

trial 0 = current value (20 mm), trial 1 = +5 %;
a secant through them lands trial 2 on 11.5 g (the relation is nearly linear);
further secant steps, then Brent once bracketed, when it is not



From the current value

Current, then +5 %; secant steps; Brent once the target is bracketed.

Both bounds

A trial at each bound (they must bracket), then Brent.

First match

Even steps from lower to upper; stop at the first value meeting $<$, $<=$, $>$ or $>=$.



The dialog (1/2)

IMAGE MISSING:

docs/decks/iterative_solve/shots/dialog.png

#	Parameter	What it means
1	Features to iterate	The features to re-run (tree order).
2	Iteration variable	The name, without #; must exist upstream.
3	Variable type	Length / Angle / Number.
4	Lower bound	Search range. Lower = upper by default: an error until you set them.
5	Upper bound	Upper end of the search range.
6	Method	Reach target / First value meeting condition.



The dialog (2/2)

IMAGE MISSING:

docs/decks/iterative_solve/shots/dialog.png

#	Parameter	What it means
7	Start from	Current value (+ First step %, 5) / Both bounds.
8	Result	Variable / Mass of created solids (+ density) / Volume of created solids; Store result as.
9	Target from variables	On: target and tolerance variables, re-read each trial. Off: typed target and tolerance.
10	Max trials	30.
11	Debug	Watch variables, feature trace, keep failed trial, run once at a value.



Outputs and messages

Published for Extract variables

Key	What it holds
#<iteration variable>	The solution.
<Store result as>	Optional: the mass (g) or volume (mm ³) at the solution.
(bodies)	The accepted trial's bodies.

Messages

	Message / what to do
Info	#x = ... after N trials; result ... Normal.
Error	The target lies beyond the lower / upper bound Widen the bounds or change the target.
Error	The bounds do not bracket the target / No value met the condition
Error	The result does not depend on the iteration variable Do the listed features read it?



Tips and limits



Set the tolerances

Length tolerance defaults to 25 mm and angle tolerance to 0 (exact).



Probe first

Debug > Run once at a value builds one trial without searching.



Downstream references

They bind to <solver>trialN<feature> and break if a later edit accepts a different trial.



Cost

Every trial re-runs the whole list.



Where it lives

Document	solvers (3cf445f157b6a28aa1d35fcf; named 'Stray_Weight_Concept_1' in Onshape)
Tabs	iterative_solve, solver_core
Tests	verified live 2026-09-22 (Stray_Weight, Design_Master derive); no test studio
Explainer	docs/explainers/solvers/solvers_explained.md
Related	Part Volume